

Ethics lecture for Design & Professional Skills

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Ethics is a big issue

- Recent example: generative AI and large language models (LLMs).
 - What issues have been raised regarding their design and use?

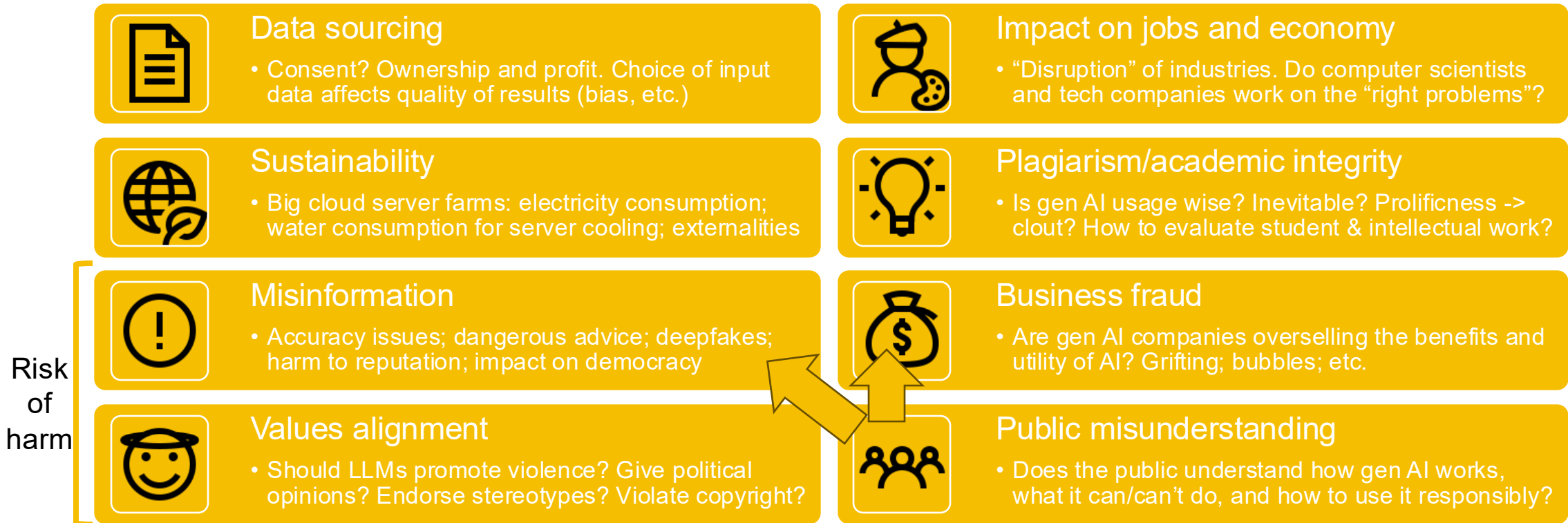
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Issues raised about gen AI



Different kinds of ethical issues

- Short-term harm to individuals.
- Long-term harm to society as a whole.
- Harm to the company developing the technology (e.g. reputational, fines/lawsuits, lost business, etc.).

Developers and tech companies make decisions

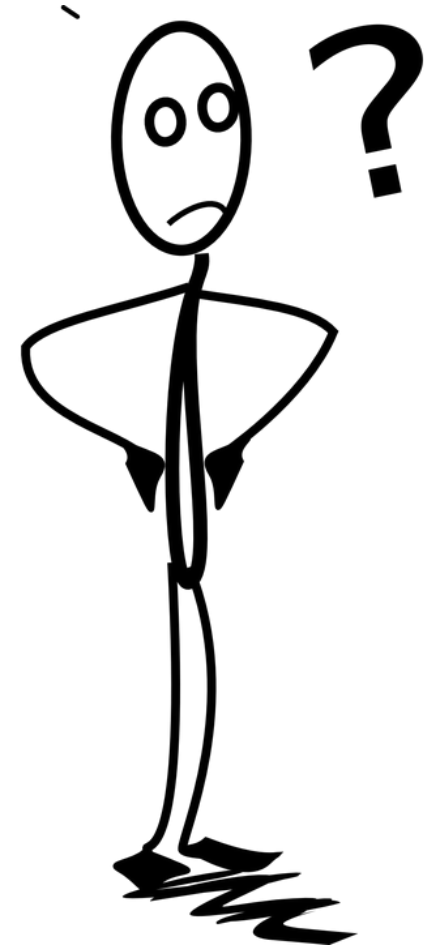
- Data sourcing: Choice of datasets affects behaviour. Ethically collecting data.
- Sustainability: Cost/benefit analysis of incorporating LLMs into business processes. Hiding environmental costs from users.
- Misinformation: Decision to deploy a technology that produces “hallucinations.” Blocking prompts that involve living public figures.
- Values alignment: Companies frequently update AI engines in response to controversy/complaints.
- Impact on jobs/economy: Choice of problems to work on determines what kind of work gets automated.
- Business fraud and public misunderstanding: Engineers have the responsibility to be honest about what tech can and can't do (expert trust issue).

Developers and tech companies make decisions

- Every choice matters: Ethical decisions occur at multiple stages of design, deployment, and use.
- Company-level decisions:
 - What datasets to use?
 - How transparent to be about environmental costs?
 - When to block harmful prompts or update models?
- Individual-level decisions:
 - Should I use AI responsibly in my work?
 - How do I balance convenience with privacy and fairness?
- Shared responsibility: Both companies and individuals shape outcomes through their choices.

Ethics is important but what is ethics?

- Difficult to talk about/teach because people disagree on what is good and right.



Plan for today

- Motivating example
- Ethics terminology and frameworks
- Examples from computer science
- Ethics canvas activity
 - Introduction
 - Small group discussion
- Big group discussion and wrap-up

Ethics is related to many concepts

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Ethics is related to many concepts

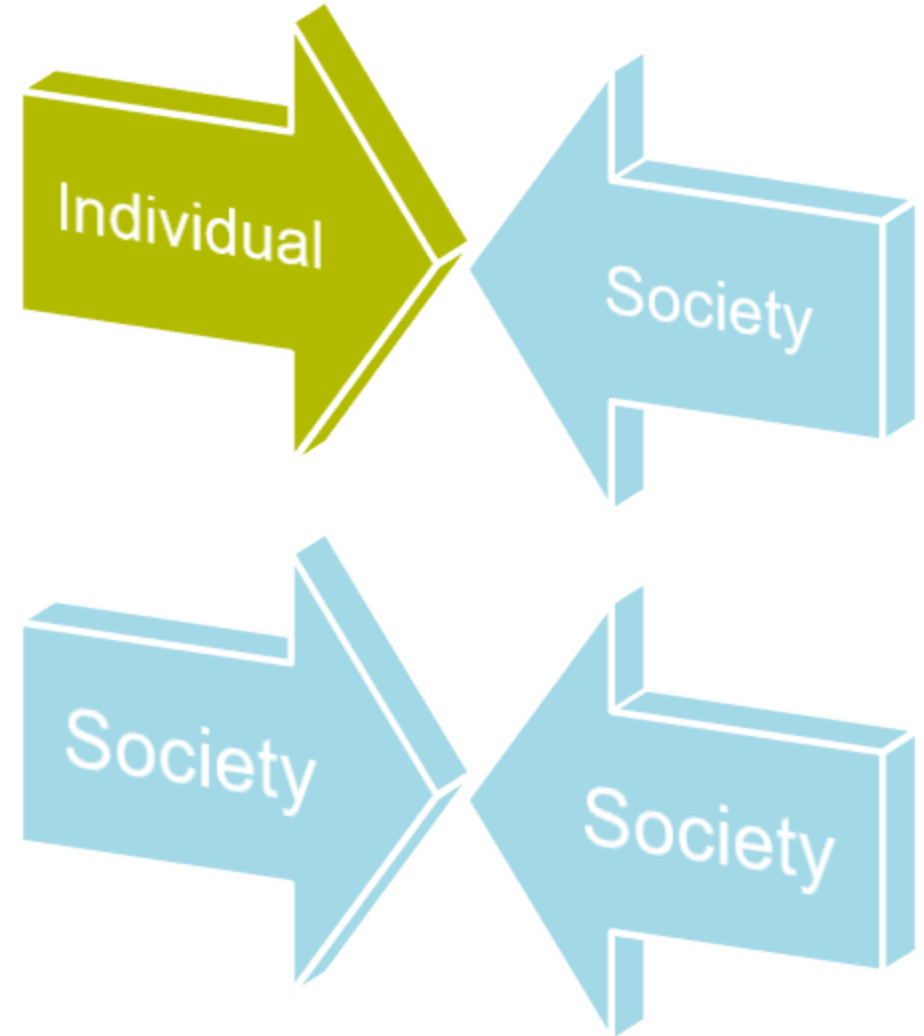


A word cloud of ethics-related concepts. The words are arranged in a roughly circular shape, with some words appearing larger and more prominent than others. The colors of the words range from light blue to purple. The words include: human rights, welfare, responsible innovation, professionalism, moral code, justice, professional ethics, social justice, safety, making the world a better place, right and wrong, principles, morality, responsibility, regulation, code of conduct, law, doing the right thing, regulation, social impact, and fairness.

human rights
welfare
responsible innovation
justice
professionalism
moral code
professional ethics
social justice
safety
making the world a better place
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principles
morality
responsibility
regulation
code of conduct
law
doing the right thing
regulation
social impact
fairness

Complications

- Ethics and the law are closely related but also frequently come apart.
- Different societies can disagree about ethics.
- A society and individuals (and organizations) can disagree about ethics:
 - Individual thinks society/org falls short of ethical standards → conscientious objection; whistleblowing.
 - Society/org thinks individual falls short of ethical standards → sanctions; punishment.
- Ethics is everyone's responsibility!

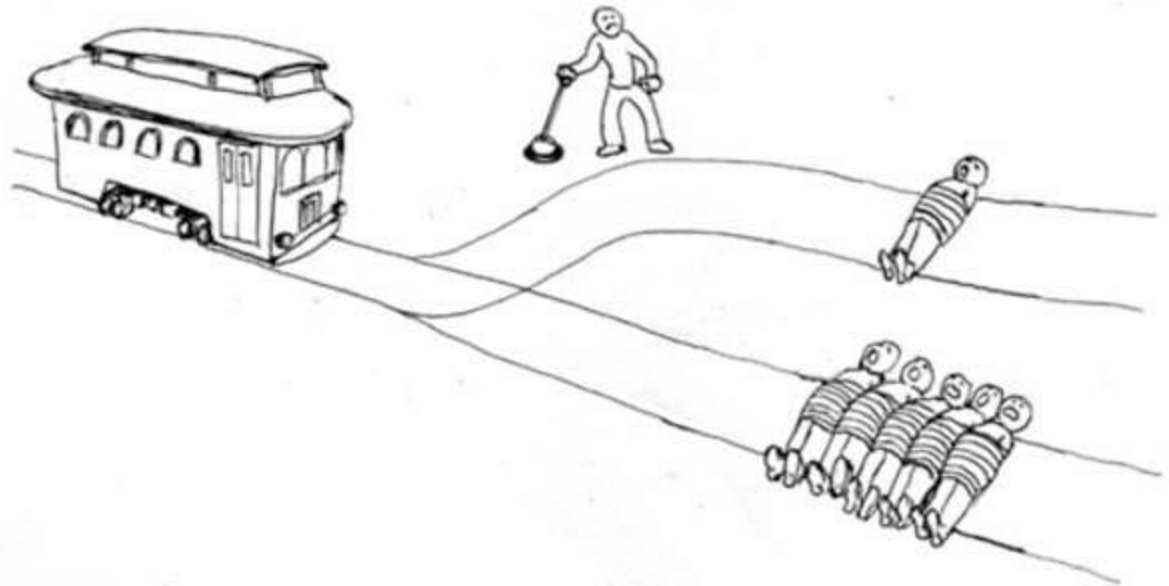


Dimensions of ethics: main frameworks

- **Deontology**
 - Ethics is about finding the right principles and rules that lead to ethical behavior.
 - Associated with Immanuel Kant but also legal systems, commandments, Golden Rule, etc.
- **Utilitarianism**
 - Ethics is about choosing actions that maximize benefit and minimize harm.
 - Associated with Jeremy Bentham but also Mozi.
- **Virtue ethics**
 - Ethics is about developing virtues and having the judgment to balance between them.
 - Associated with Aristotle and Confucianism.
- **Care ethics**
 - Ethics is about honoring relationships of care between humans (in between utilitarianism and virtue ethics?).
 - Associated with Carol Gilligan and Nel Noddings; feminist theory of ethics.

Deontology vs. consequentialism

- Trolley problems as a way to understand people's moral intuitions regarding deontology vs. consequentialism.
- Applications to self-driving cars (see e.g. <https://www.moralmachine.net/>)



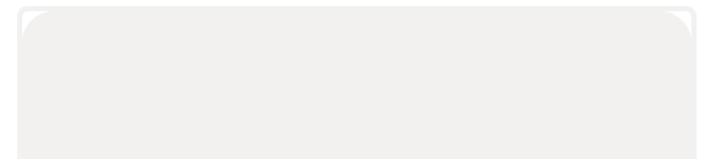
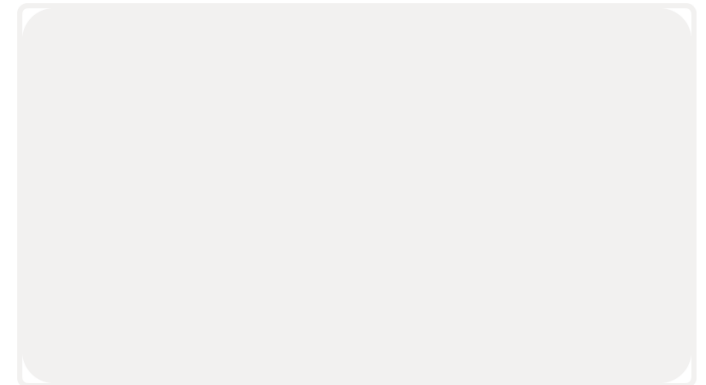


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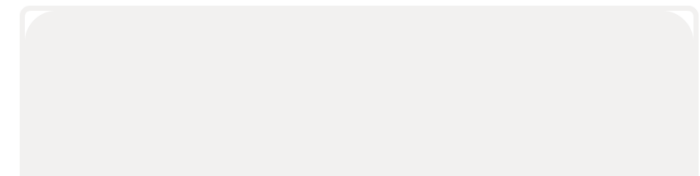
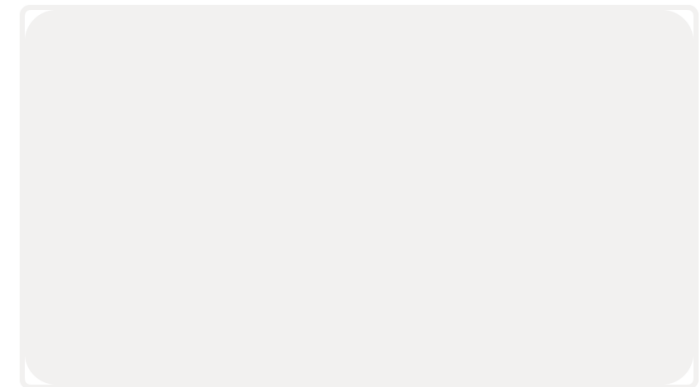


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Dimensions of ethics: motivation for doing ethics

- **Social responsibility:** civic duty as a member of society that cares about the effects of one's actions on that society.
- **Professional responsibility:** part of being a good/professional engineer – consider the responsibilities you hold to your employer, clients, etc. who rely on your work. Protect your company or engineers as a class from financial or reputational harm (distrust).
- **Responsibility to self:** expression of/staying true to your moral code; making sure your creations are embedded with your values.

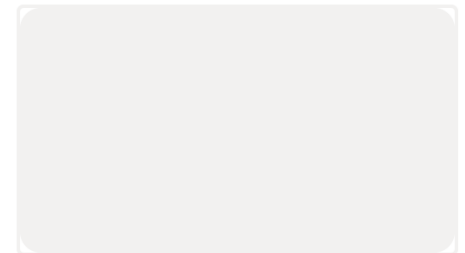
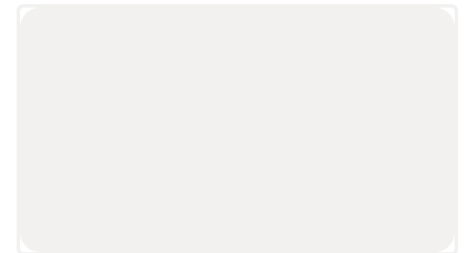
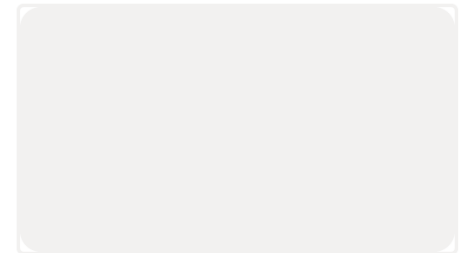


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Dimensions of ethics: areas of ethics

- **Hurt/harm** (and conversely, beneficence/benefit).
- **Fairness.**
- **Autonomy.**
 - Includes free speech and other personal liberties, workers' rights, etc.
- **Privacy.**
- **Sustainability.**
- (Not an exhaustive list.)

Dimensions of ethics: CS ethics

- **Data ethics:** Ethical collection and use of data appropriate for task.
 - Related to harm and privacy.
- **Algorithm ethics:** Ethical behavior by computer systems involved in high-stakes decision making.
 - Related to harm and fairness.
- **Interaction ethics:** Ensuring computer systems can be trusted by humans and other systems and have positive effect on society.
 - Related to harm, autonomy, privacy, and security.

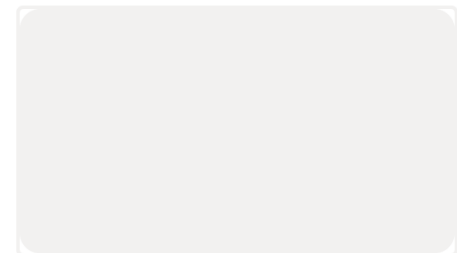
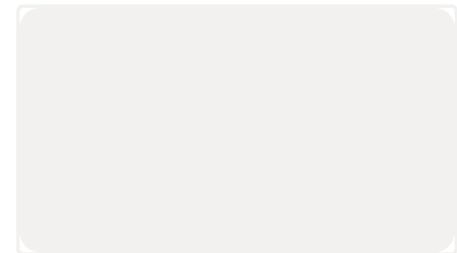
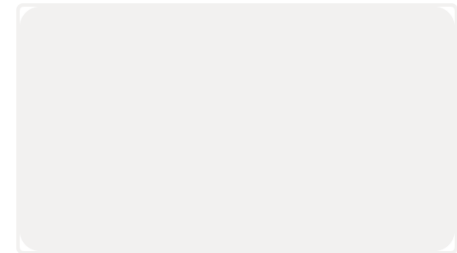


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Ethics examples from Computer Science

Data ethics: example

- Increasing reliance on data collected through websites for advertising has led to an abuse of user data, non-consensual sharing, data used to manipulate users, etc.
- Before GDPR disclosure requirements, many users unaware of the extent of data collection and how it was being used.



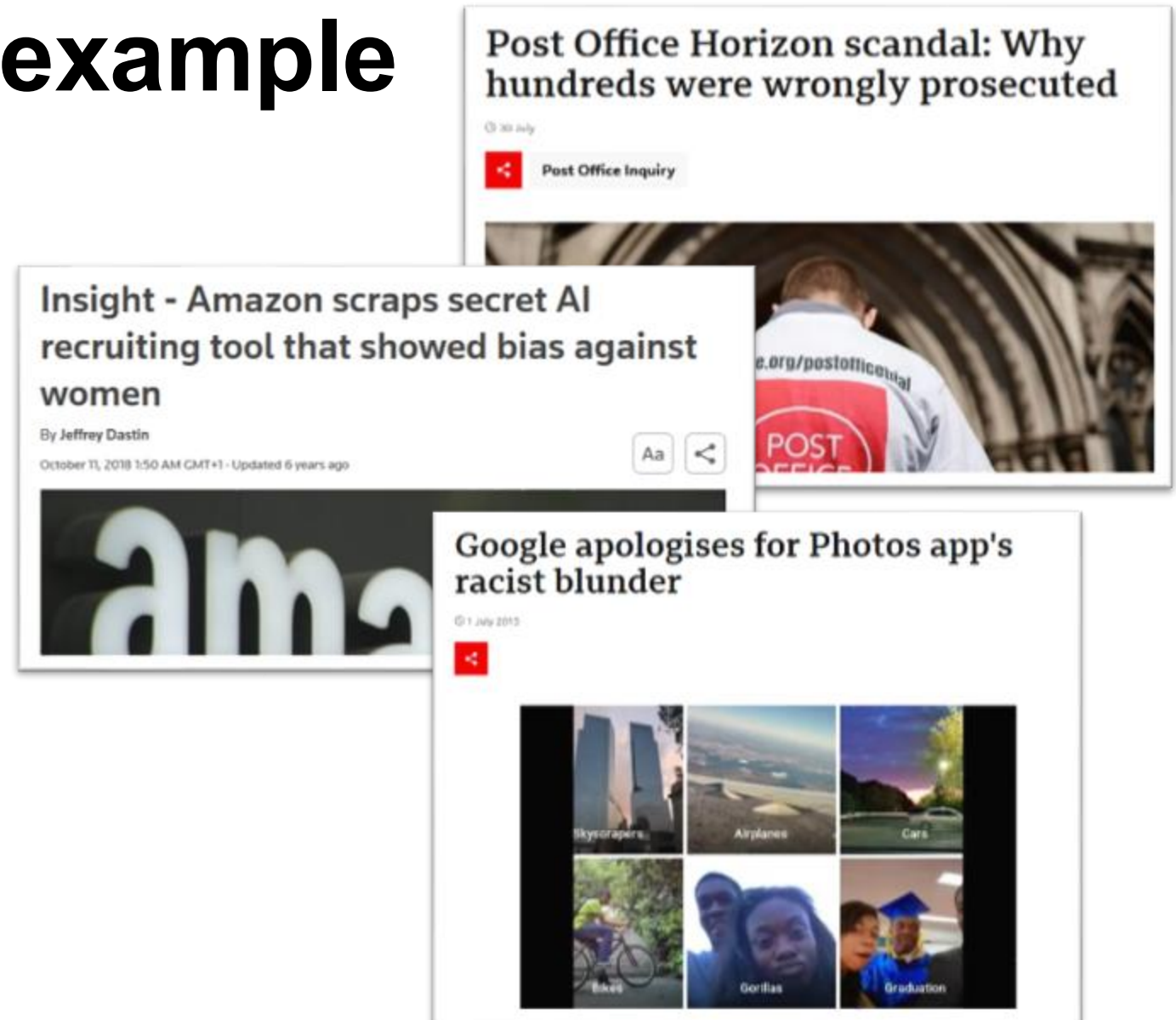
Data ethics: GDPR regulation

- Principles for collecting and storing data to minimize harm to the owners of the data.



Algorithm ethics: example

- Biased or faulty algorithms can cause harm to people.
- Real-life consequences can be severe.
- Deployment of software not thoroughly tested can be costly.



Algorithm ethics: regulation

- Discriminatory algorithms may violate the Equality Act 2010, if making decisions related to employment and pay.

Interaction ethics: example

- Do/can people trust technology?
- Does technology make people better or worse?

The Looming Threat of Unsecured IoT Devices: A Deep Dive



Author: Zechariah Oluleke Akinpelu
Date Published: 10 July 2024
Read Time: 6 minutes

SHARE     

YouTube's algorithms might radicalise people – but the real problem is we've no idea how they work

Published: January 21, 2020 2:35pm GMT

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Does YouTube create extremists? A [recent study](#) caused arguments among scientists by arguing that the algorithms that power the site don't help radicalise people by recommending ever more extreme videos, as [has been suggested](#) in recent years



Case study: dark design patterns

- UI elements designed in order to manipulate user, often to increase profits of the company.
- <https://www.deceptive.design/types>

Examples	
Confirmshaming	Hidden subscription
Disguised ads	Obstruction
Fake scarcity	Preselection
Fake urgency	Trick wording
Hard to cancel	Visual interference

Manipulative elements of social media sites

- More subtle and less serious/fraudulent, but still manipulative.
- <https://purl.stanford.edu/rq188wb9000>

Examples	
Bright red notifications that user wants to remove by clicking	Disabling or hiding ability of user to control page contents
Variable schedule of "rewards" in the form of likes / comments / social approval	Endless scroll to keep users on the site (ad-centered attention economy)
Placement of ads and paid site features in prime page locations	Using friends' behavior to prompt user to take similar action (social pressuring)

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Which two of the following are examples of manipulative elements of social media sites discussed in the presentation?

0 ✓

Bright red notifications that the user wants to remove by clicking

0 ✗

Using data analytics to improve user experience

0 ✓

Disabling or hiding the user's ability to control page content

0 ✗

Employing AI-powered chatbots to provide customer support



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What would you do?

Option	Count
Do nothing	10
Pull the lever	21

Ethics Canvas activity

Digital Ethics Canvas

The Digital Ethics Canvas is designed to help you **identify ethical risks with your software.**

When to use it?

- **Design stage:** Use the canvas early to explore options and assess ethical risks when defining solution features, enabling better mitigation.
- **Development stage:** Apply the canvas to evaluate ethical risks in implementation choices and their impact on users and the environment.
- **Use time:** Users can use the canvas to compare tools and select based on ethical criteria.

DIGITAL ETHICS CANVAS

CONTEXT	SOLUTION	BENEFITS
WELFARE RISK - Can the solution be used in harmful ways, in particular with regards to vulnerable populations? - What kind of impacts can arise from the solution base? - What type of protection does the solution have against attacks or misuse? MITIGATION		
FAIRNESS RISK - How accessible is the solution? - What kinds of biases may affect the results? - Can the outcomes of the solution be different for different users or groups? - Could the solution contribute to discrimination against people or groups? MITIGATION		
AUTONOMY RISK - Can users understand how the solution works and what its limits are? - Can users able to make informed choices, including in their use of the solution and tool? - How does the solution affect user autonomy and agency? MITIGATION		
PRIVACY RISK - What data does the solution collect? - Is it collecting personal or sensitive data? - Who has access to the data? - How is the data protected? - Could the solution disclose / be used to disclose private information? MITIGATION		
SUSTAINABILITY RISK - What is the carbon footprint of the solution? - What types of resources does it consume (e.g. water) and produce (e.g. waste)? - What type of harmful value is involved? MITIGATION		

This canvas is licensed under CC BY-NC-SA. Copyright © 2021 UCL. All rights reserved. For more information, please visit: <https://www.ucl.ac.uk/digital-ethics>

Digital Ethics Canvas

Key steps

- Provide a brief overview of the context and proposed solution.
- Outline the expected benefits.
- Identify risks through the lenses of Welfare (harm), Fairness, Autonomy, Privacy, and Sustainability.
- Detail mitigation strategies for each identified risk.
- Remember! The Canvas is iterative!

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SUSTAINABILITY RISK - What is the carbon footprint of the solution? - What types of resources does it consume (e.g. energy) and produce (e.g. waste)? - What type of human labour is involved? MITIGATION		

Digital Ethics Canvas

- State the **expected** benefits of the digital solution, focusing on the needs it addresses and the target users.
- "What needs is the solution intended to meet and for whom?"

CONTEXT

SOLUTION

BENEFITS

- Record relevant contextual details in the “Context” zone – where, when, how?
- Document elements of the "Solution" being evaluated e.g. web interface, ML model etc.

[illegible]

Digital Ethics Canvas

Welfare:

- **Risk:** Could the solution be used harmfully, especially impacting vulnerable groups, or cause negative effects due to errors?
- **Mitigation:** What safeguards and error-handling mechanisms can be implemented to protect users and prevent misuse?

WELFARE

RISK

- Can the solution be used in harmful ways, in particular with regards to vulnerable populations?
- What kind of impacts can errors from the solution have?
- What type of protection does the solution have against attacks or misuse?



MITIGATION



Digital Ethics Canvas

Fairness:

- **Risk:** Could biases or discrimination arise, leading to unequal outcomes or reduced accessibility?
- **Mitigation:** What measures, like bias-checking tools or evaluations, can ensure fairness and inclusivity?

FAIRNESS

RISK

- How accessible is the solution?
- What kinds of biases may affect the results?
- Can the outcomes of the solution be different for different users or groups?
- Could the solution contribute to discrimination against people or groups?



MITIGATION



Digital Ethics Canvas

Autonomy:

- **Risk:** Can users understand how the solution works, its limits, and how it affects their autonomy?
- **Mitigation:** What measures ensure users can make informed choices and maintain agency in using the solution?

AUTONOMY

RISK

- Can users understand how the solution works and what its limits are?
- Are users able to make choices (e.g. consent, settings) in their use of the solution and how?
- How does the solution affect user autonomy and agency?



MITIGATION



Digital Ethics Canvas

Privacy:

- **Risk:** What data does the solution collect, including personal or sensitive information, and how might private information be disclosed?
- **Mitigation:** What protections are in place to safeguard data and limit unauthorized access or disclosure?

PRIVACY

RISK

- What data does the solution collect
- Is it collecting personal or sensitive data
- Who has access to the data?
- How is the data protected?
- Could the solution disclose / be used to disclose private information?



MITIGATION



Digital Ethics Canvas

Sustainability:

- **Risk:** What is the solution's carbon footprint, resource consumption, waste production? Is there any form of labour exploitation occurring?
- **Mitigation:** What practices can minimize environmental impact and promote sustainable resource use?

SUSTAINABILITY

RISK

- What is the carbon footprint of the solution?
- What types of resources does it consume (e.g. water) -and produce (e.g. waste)?
- What type of human labor is involved?



MITIGATION



Activity Time!

Please scan the QR code or click the link to reach the Miro board



[Link to Miro Board](#)

Wrap-up

Advocating for ethics: tools in your toolbox

- Expanding spheres of action:
 - Design decisions.
 - Testing and analysis, code auditing – is your code/system energy-efficient, secure, and bug-free?
 - Transparency and public education/communication.
 - Advocacy (by oneself or in groups)
 - Legislation and regulation.



Google's “Don't be evil” motto

- Not included in Alphabet's Code of Conduct since Alphabet's creation, and removed from the preface of Google's Code of Conduct in 2018.
- What behavior would it make you avoid? What does it mean to remove it?

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Google Code of Conduct

Preface

"Don't be evil." Googlers generally apply those words to how we serve our users. But "Don't be evil" is much more than that. Yes, it's about providing our users unbiased access to information, focusing on their needs and giving them the best products and services that we can. But it's also about doing the right thing more generally – following the law, acting honorably and treating each other with respect.

The Google Code of Conduct is one of the ways we put "Don't be evil" into practice. It's built around the recognition that everything we do in connection with our work at Google will be, and should be, measured against the highest possible standards of ethical business conduct. We set the bar that high for practical as well as aspirational reasons: Our commitment to the highest standards helps us hire great people, build great products, and attract loyal users. Trust and mutual respect among employees and users are the foundation of our success, and they


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Google Code of Conduct

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VIII. Conclusion

Google aspires to be a different kind of company. It's impossible to spell out every possible ethical scenario we might face. Instead, we rely on one another's good judgment to uphold a high standard of integrity for ourselves and our company. We expect all Googlers to be guided by both the letter and the spirit of this Code. Sometimes, identifying the right thing to do isn't an easy call. If you aren't sure, don't be afraid to ask questions of your manager, Legal or Ethics & Business Integrity.

And remember... don't be evil, and if you see something that you think isn't right – speak up!

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